

Cognate Facilitation Effect on Sentence Comprehension: What's the mechanism?

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Bilingual Cross-language Activation

- Languages are activated in parallel (e.g., Altenberg & Cairns, 1983; Beavillian & Grainger, 1987; Brysbaert et al., 1999; De Groot, et al., 2000; Dijkstra et al., 1999; Jared & Kroll, 2001; Van Heuven et al., 1998; Van Hell & Dijkstra, 2002)
- Observed in reading, listening, and speaking (Dijkstra, 2005; Kroll et al., 2006; Marian & Spivey, 2003).
- Faster processing of cognates provides evidence of parallel activation of words in bilinguals (Kroll et al., 2014).

Cognates between English & Spanish

Identical	
Piano	Piano
Hospital	Hospital
Chocolate	Chocolate

Non-Identical	
System	Sistema
Machine	Máquina
Cognate	Cognado

Cognate Facilitation Effect

Demonstrated in a variety of tasks

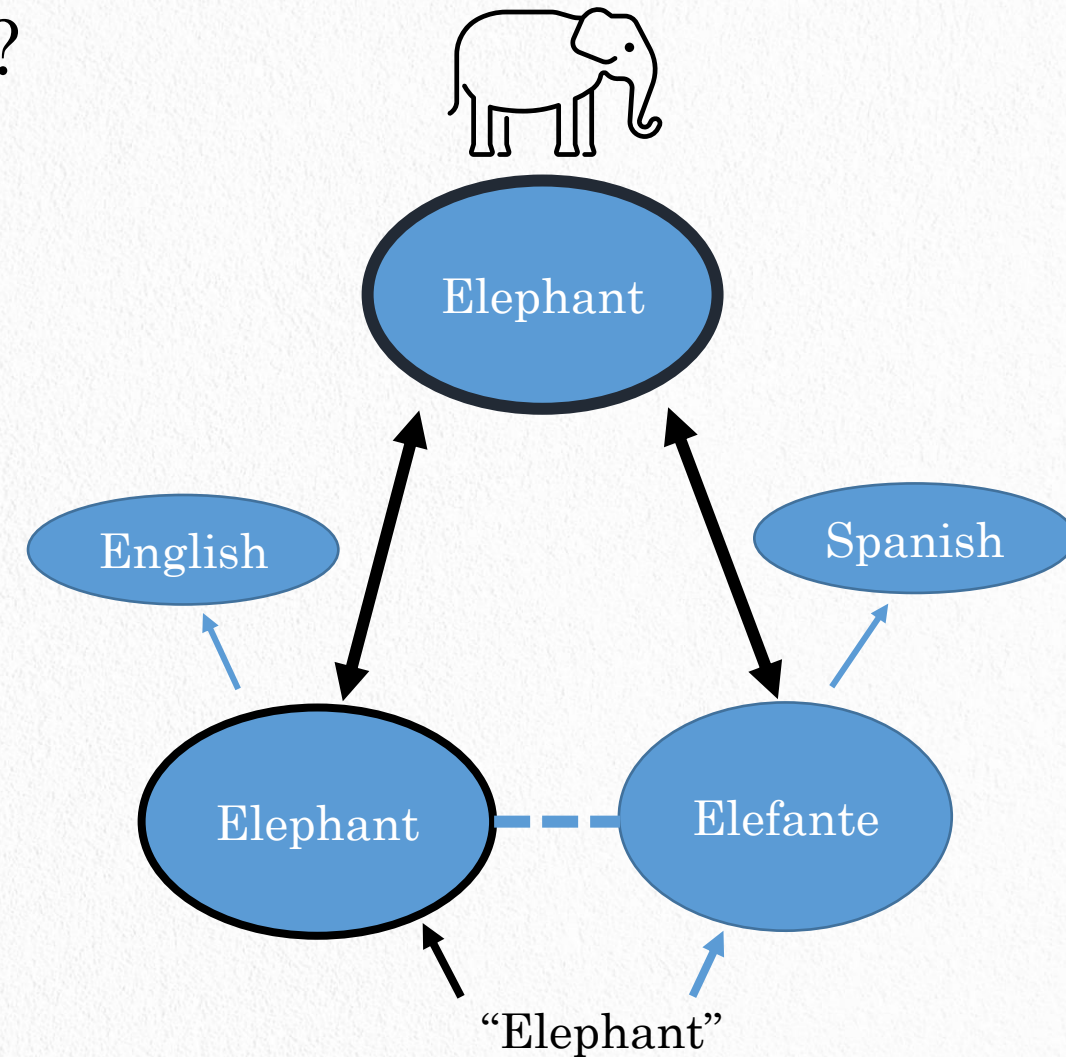
- Visual lexical decision task
 - Word production
 - Word association
-
- Inhibitory effect in word recognition and production if cognates are non-identical
(Poort, & Rood, 2017)

(e.g., Altarriba et al, 1996; Duyck et al, 2007; Schwartz & Kroll, 2006; Libben & Titone, 2009; Van Hell and de Groot, 2008; Schwartz & Kroll, 2006; Van Hell & de Groot, 2008; Libben & Titone, 2009; Dijkstra et al., 1998; De Groot et al., 1994; Kroll & Stewart, 1994; Kroll et al., 1999; Van Hell & Dijkstra, 2002).

Why Are Cognates Recognized Faster?

Bilingual Interactive Activation Model Plus (BIA+)

- Information from two languages is represented in an integrated system
- Not language selective



(Dijkstra & Van Heuven, 2002)

Beyond Cognate Word Recognition

- Does facilitated recognition of cognates influence later stages of processing?
- Lauro & Schwartz (2019)
 - Hypothesis: may facilitate integration of larger units of meaning
 - Eye-tracking study
 - Manipulated cognate status of the first two nouns of a sentence
 - Would there be facilitated processing of later-occurring pronoun?

Lauro & Schwartz (2019)

Noncognate Nouns

The horses feared the animal near the barn, but it was harmless.

Cognate Nouns

The guitar was louder than the drums, so it was put at a lower volume.

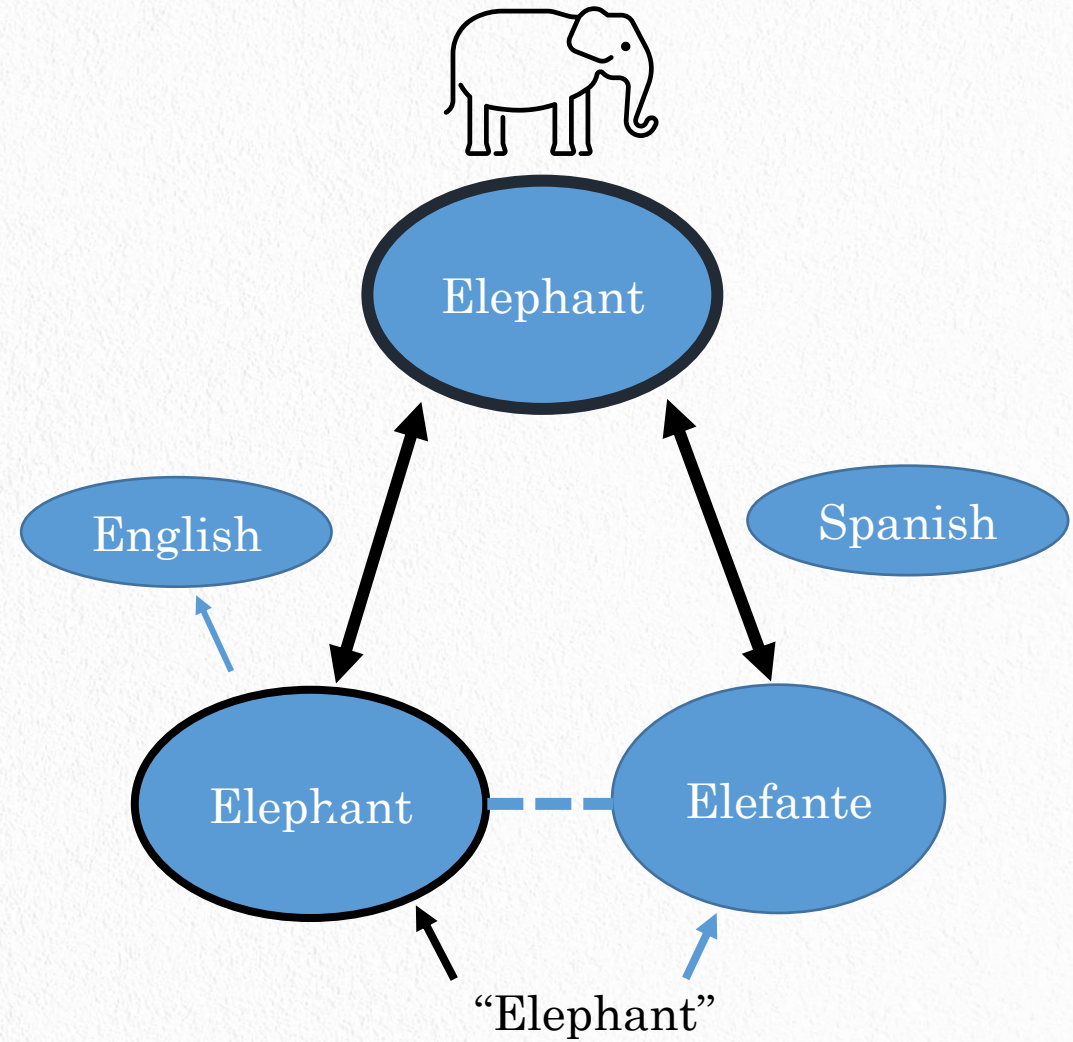


Results: Faster reading times when one of the nouns was a cognate

Why Did Cognates Facilitate Later Stages of Processing?

Possible Explanations

1. Faster access to meaning facilitates computations involved in sentence comprehension



Why Did Cognates Facilitate Later Stages of Processing?

Possible Explanations

1. Faster access to meaning facilitates computations involved in sentence comprehension.
2. The memory trace of cognates is stronger in working memory because of their high activation.



The guitar was louder than the drums, so it was put at a lower volume.

Working Memory

- Recruited for complex behaviors such as reasoning and problem solving
- Working memory is crucial for sentence comprehension
 1. Holding - keep relevant information active in memory
 2. Computation - processing of additional information

The guitar was louder than the drums, so it was put at a lower volume.

Reading Span Task

- Participants read sentences and verify the plausibility of the sentences while simultaneously recalling words at the end of each set.
 - Performance modulated by the difficulty on computing the sentence
- Set size can range from 2-6
- Different methods of scoring

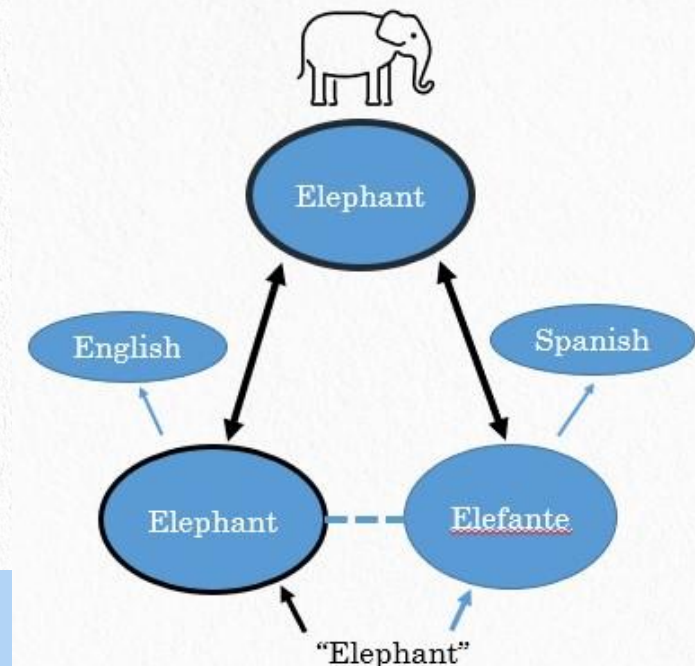
*It was the purse that snatched the thief.	Tree
*The red barn jumped the fence besides the horses.	Attempt
The food that Nathan prepared poisoned Susan.	Pencil
The woman thanked the girl that found the wallet.	Forest
*It was the sleeping couple that woke the loud noise.	Devil

* *Implausible sentence*

(Conway et al., 2005)

Pilot Study

- Research aim:
 - Test hypothesis
 - faster semantic access of cognates facilitates computation of sentence meaning.
- Prediction:
 - Participants should recall more words when sentences compromise of cognates relative to when sentences have no cognates



Pilot Study

- Reading Span Task
- Self-paced
- Zoom & Qualtrics
- N=20
- Set sizes
 - Two sets of 3, 4, 5, and 6 sentences

Stimuli

- 2 sets of 36 sentences
 - Cognate content words
 - No cognates

Sentence presented in the middle of screen

- 6000ms to make decision

Follow-up word for recall

- Presented for 2000ms

Cognate version

It was the **criminal** that
escaped the **prison**.

YES

QUARREL

It was the **virus** who
studied the **scientist**.

NO

SONG

It was the **altar** that
was **presented** as a
sacrifice.

NO

BEAUTY



Recall:
Quarrel
Song
beauty

Non-Cognate version

It was the singer that
performed on the stage

YES

SPIDER

It was the lily that sat
on the toad

NO

WINDOW

It was the movie that
watched the
housekeeper

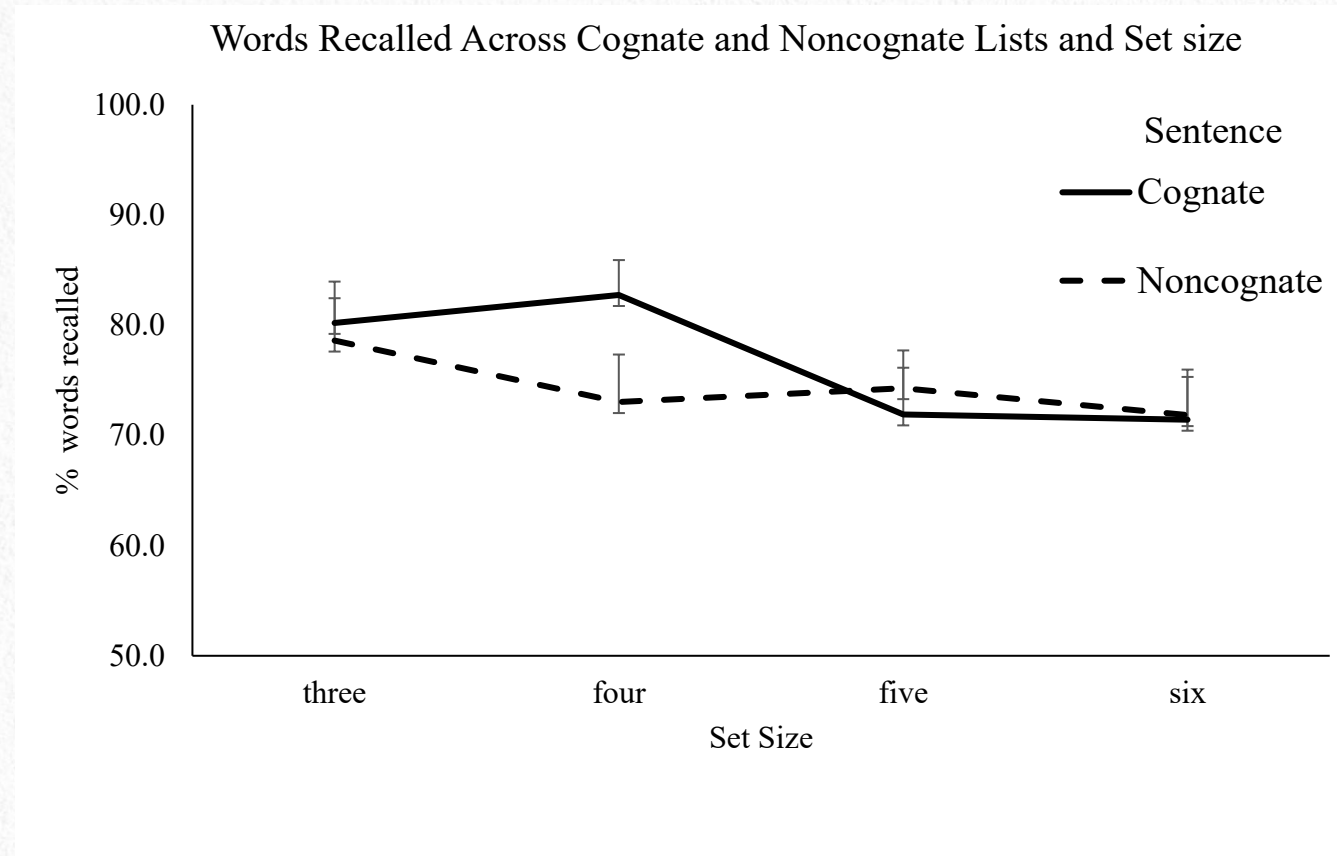
NO

FOREST



Recall:
Spider
Window
Forest

Results



Present Study

- Replication
- Larger sample size
- More controlled experiment
 - Last word of the sentence is the target recall word
 - Controlled sentence length and structure

Participants

- N=60
- Woodcock-Muñoz Language Survey-Revised^[1]
 - Picture Vocabulary & Verbal Analogies (in both languages)
- ESPADA Questionnaire^[2]
 - Frequency of usage, exposure, context of use, absolute and relative ratings.

Language Proficiency

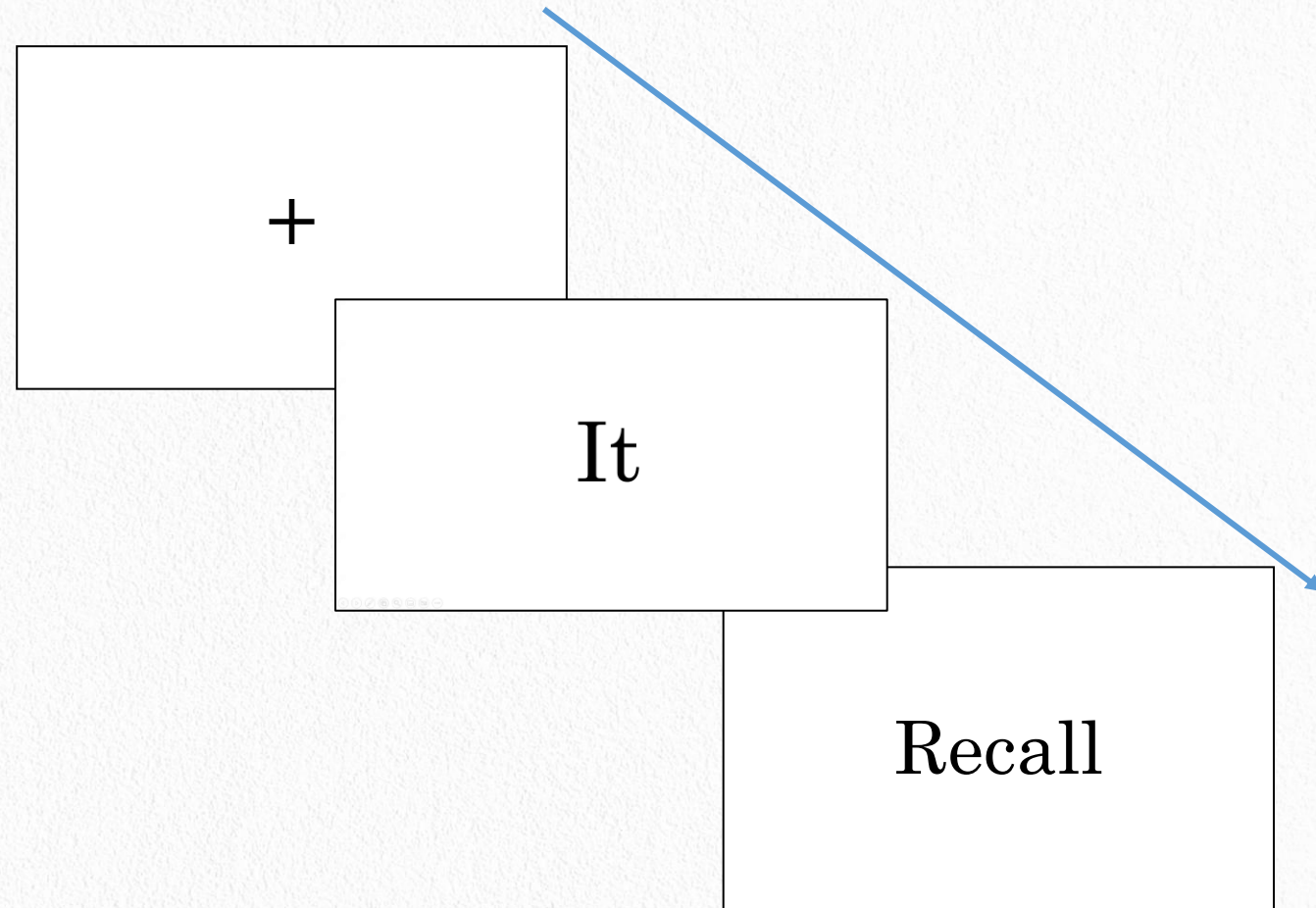
	English	Spanish
Age of Acquisition	6.53	1.40
Self-Rating Proficiency		
Reading	3.60	
Years of Education	11.52	4.85
WMLS-R		
Picture Vocabulary	502.63	504.35
Verbal Analogies	507.12	507.46

[1] (Woodcock et al., 2005)

[2] (Francis & Strobach, 2013)

Method- Reading Span Task

1. Computation Component:
Reading sentences and deciding plausibility
 2. Holding Component:
Recall target words within that specific set size.
Target words= last word of a sentence
- Rapid Serial Visual Presentation (RSVP)
 - 250ms
 - 3000ms word-to-be-recalled



Stimuli

- Set Size (3-6)
- Plausibility of Sentence (plausible/implausible)
- Cognate Status of Sentence (cognate/noncognate)

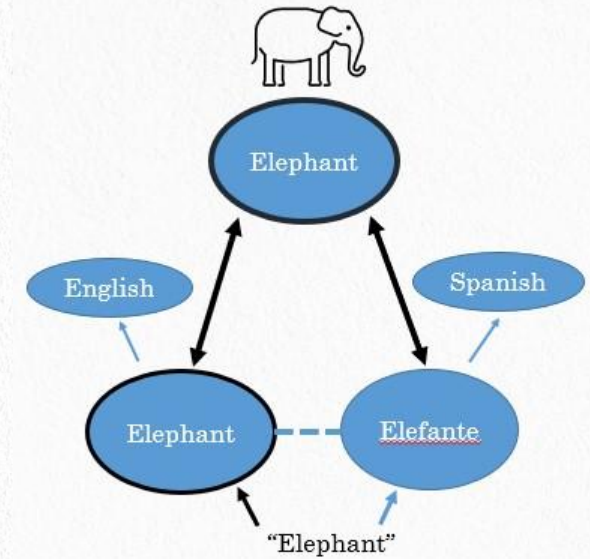
	Cognate	Noncognate
Plausible	It was the tourist at the hotel who ordered a taxi .	It was the party that Julia hosted that woke the neighbors.
Implausible	It was the hotel at the tourist that ordered a taxi .	It was the neighbors that Julia hosted that woke the party.

Procedure

- 1) Picture naming task of Woodcock-Muñoz Language Survey-Revised
- 2) Reading Span task
- 3) Verbal analogies task of Woodcock-Muñoz Language Survey-Revised
- 4) Reading Span task
- 5) Complete ESPADA survey

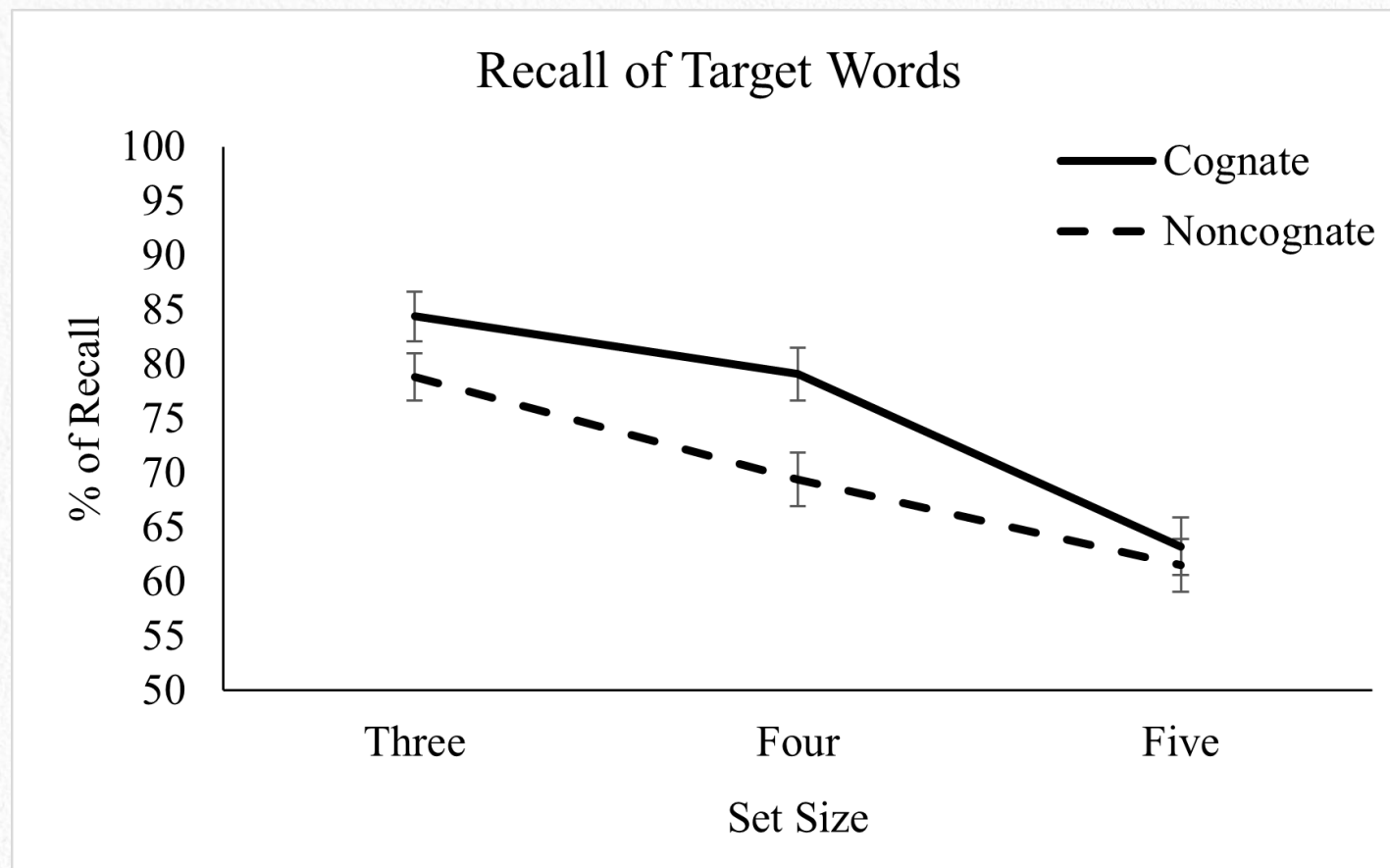
Reading Span tasks

- Test computation hypothesis
 - Cognate status manipulation of words in the sentences
- Test memory trace hypothesis
 - Cognate status manipulation in target words only



The guitar was louder than the drums, so it was put at a lower volume.

Results



Discussion

- There was a higher target word recall for sentences composed of cognates, specifically for set size 4
- Supporting hypothesis that cognates facilitate the computation of meaning of sentences
- Second hypothesis: Strength of memory trace
 - Can the effect be due to cognates having a stronger memory trace because of their high activation?



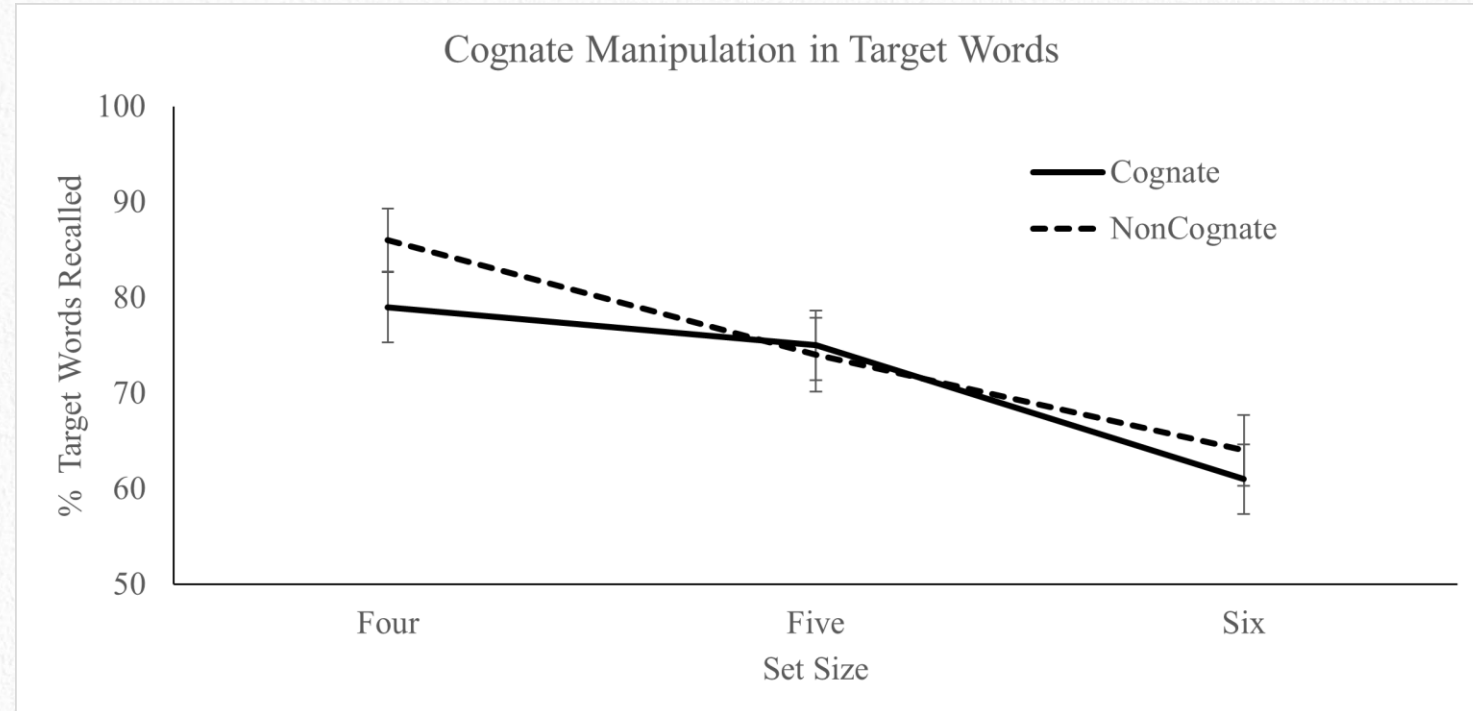
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What About Strength of Memory?

Preliminary
Data (N=35)

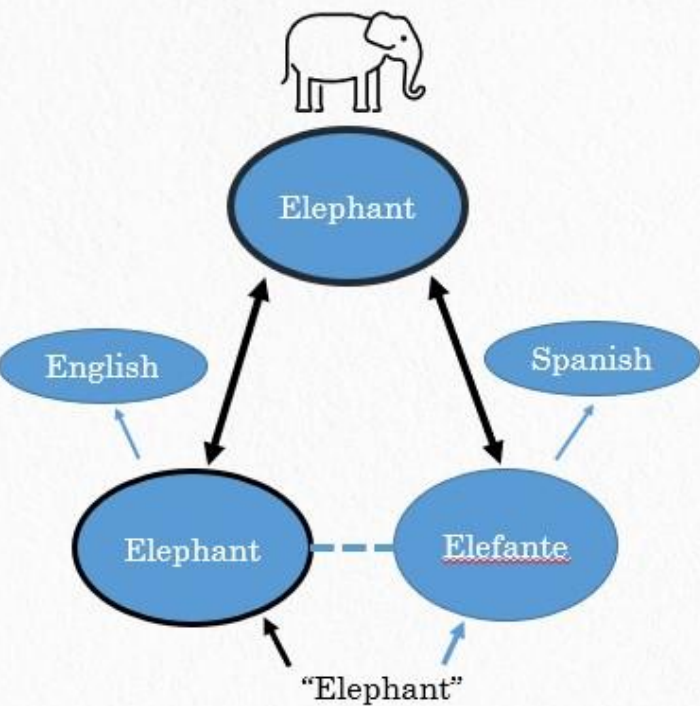
-Cognate status
manipulation in
word-to-be
recalled

- No effect of
cognates

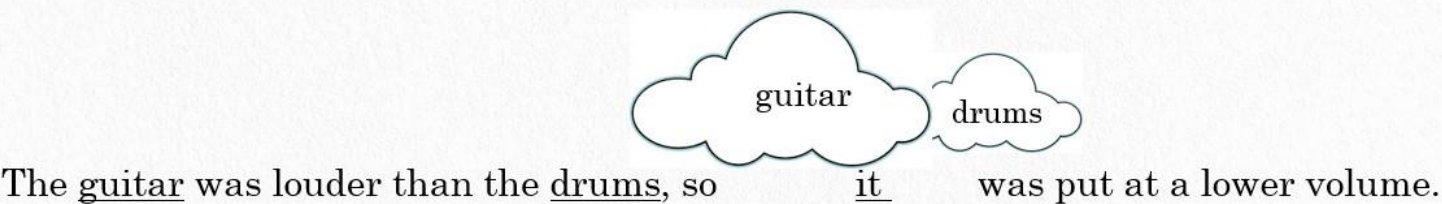


The guitar was louder than the drums, so it was put at a lower volume.

Conclusion



Computation



Strength of memory trace

Thank you!



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